Velagapudi Sai Tanmai

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Skills

Languages: C++, C, Java, Python
Tools/Platforms Git and GitHub, VS Code, PostgreSQL, Docker, Excel, Power BI
Data Science & ML: Numpy, Pandas, Matplotlib, Seaborn, Scikit-Learn
Soft Skills: Problem-Solving, Team Player, Leadership, Adaptability

Projects

PyroLens – Real-Time Global Forest Fire Monitoring Platform | GitHub August 2025

- Developed a real-time global fire monitoring system using FastAPI, PostgreSQL + PostGIS, and enabling live visualization of forest fire hotspots on an interactive world map.
- Built an automated 10-minute-interval ETL + geospatial analytics pipeline that ingests NASA FIRMS data, cleans and updates a PostGIS database, and applies DBSCAN clustering with centroids and convex hulls to identify high-risk fire zones.
- Implemented geospatial analytics using DBSCAN clustering, centroids, and convex hulls to identify and visualize high-risk fire zones.

Tech: PostgreSQL/PostGIS, Pandas, Nasa Firms API, FastAPI, Python.

Alchemy of SMILES– Generative AI for Noval Drug Discovery | GitHub April – May 2025

- Engineered a generative Char-RNN (LSTM) model to synthesize novel drug-like molecular structures (SMILES), trained onacurated dataset of bioactive compounds targeting EGFRfrom theChEMBL database.
- Designed a robust cheminformatics pipeline using RDKit to preprocess data based on Lipinski's Rule of 5 and evaluate generated sequences for chemical validity, uniqueness, and drug-likeness (QED).
- Analyzed and visualized molecular generation performance and property distributions to validate model effectiveness.

Tech: Python, PyTorch, RDKit, Pandas, Matplotlib, GoogleColab

Training

LPU (Centre for Professional Enhancement)

Mastering Data Structures and Algorithms: July 2025

- Completed Data Structures & Algorithms training at LPU, covering core concepts and problem-solving strategies.
- Practiced implementation of key structures such as arrays, linked lists, stacks, queues, trees, and graphs on various platforms like leetcode, GeekforGeeks and Code360.
- Strengthened algorithmic thinking through exercises in recursion, sorting, searching, and time–space complexity.

Tech: C++, Python, Data Structures & Algorithms, LeetCode, GeeksforGeeks, Code360

Certificates

Privacy and Security in Online Social Media - NPTEL October 2025
Master Generative AI & Generative AI tools - Infosys Springboard September 2025
Computer Communications Specialization - Coursera March 2024

Achievements

- **Achieved Top 50 in the national-level Cyber Security Innovation Challenge 1.0** February 2026
  In the DSCI Government Hackathon focused on predictive analytics; currently advancing through Stage 2
- **Secured 8th Position among 200+ Teams** November 2025
  In a university-level hackathon focused on innovation & problem-solving

Education

Lovely Professional University Punjab, India
Bachelor of Technology - Computer Science and Engineering; ***CGPA: 7.77*** *August 2023 - Present*

Sri Chaitanya Intermediate College Vijayawada, Andhra Pradesh
Intermediate; ***Percentage: 90%*** *April 2021 - March 2023*

Sri Chaitanya Olympiad School Vijayawada, Andhra Pradesh
Matriculation; ***Percentage: 75%*** *April 2020 - March 2021*